

Gradient methods find the causal rows; the true provenance labels do not

Each bar: how well one row-scoring method predicts the measured effect of deleting its chosen rows (rank correlation over 10 delete-and-retrain runs). Selected for the pipeline: L3_defif_c10 ($\rho = 0.867$). Thresholds preregistered: $\rho \geq 0.5$ validated; $\rho < 0.2$ fails; selection = best eligible primary ρ , contenders within 0.05 tie-broken by cross-seed rank correlation

